Short Analysis

Decoding Comparison:

Greedy: Always picks the highest-probability token. It is fully deterministic but prone to repetitive loops.

Temperature: Scales logits before softmax.Another thing is Low temperature (0.3) is safe and repetitive; high temperature (1.8) increases creativity and randomness.

Top k: Restricts sampling to the top k candidates.
Small k (3) ensures safety large k (20) increases diversity.

Coherency vs. Diversity:

Most Coherent: Greedy decoding and Low Temperature (0.3) provide the best local syntactic structure, though they often get stuck in repetition loops.

Most Diverse: High Temperature (1.8) and Wide Top k (20) produce the most diverse text by breaking out of repetitive patterns.